

CHANDRA MOULI DASARI

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PROFESSIONAL SUMMARY

M.S. Computer Science candidate at UF (GPA 3.79/4.0) with 2 years of industry experience building ML pipelines and data systems at scale. Seeking a Research Assistant position in Dr. Karanth's lab to apply machine learning and causal inference methods to cancer epidemiology and health disparities research. Experienced in survival analysis, NLP, and working with large structured health datasets. Motivated by the intersection of computational methods and equitable public health outcomes — particularly in lung and oropharyngeal cancer survivorship.

TECHNICAL SKILLS

ML / AI: Survival Analysis (Cox PH, DeepSurv, lifelines), XGBoost, scikit-learn, PyTorch, TensorFlow, CNNs, NLP (TF-IDF, NER), Causal Inference (DoWhy), Risk Modeling, LLM Evaluation

Health Data: SEER Database, NHIS, Real-World Data Pipelines, HIPAA-Compliant Data Handling, Epidemiological Dataset Processing

Data Engineering: Python (Pandas, NumPy, SciPy), ETL Pipelines, PostgreSQL, PL/pgSQL, MongoDB, CI/CD, Git, Linux

Visualization: Matplotlib, Seaborn, Streamlit, Interactive Dashboards, Reproducible Workflows

Cloud: AWS (EC2, S3), Scalable Microservices, Version-Controlled Deployments

RESEARCH-TARGETED & ACADEMIC PROJECTS

Autonomous AI Data Scientist Agent | Python, LLMs, LangChain, Scikit-learn, SHAP, Streamlit

- ▶ Engineered an AI-powered data science assistant capable of automating exploratory data analysis, feature engineering, model selection, and performance evaluation from raw datasets.
- ▶ Integrated LLM-based reasoning with Python ML pipelines to generate automated insights, statistical summaries, visualizations, and explainable model recommendations.
- ▶ Implemented automated ML workflows using XGBoost, Random Forest, and SHAP-based interpretability, reducing manual experimentation and analysis time.
- ▶ Developed an interactive Streamlit platform enabling users to upload datasets and receive end-to-end ML reports with reproducible outputs.

Real-Time Fraud Detection & ML Monitoring Platform | Kafka, Spark, FastAPI, MLflow, Docker

- ▶ Designed and deployed a real-time machine learning pipeline for transaction anomaly detection using event-driven data streaming architecture.
- ▶ Built Kafka-based ingestion workflows, automated feature engineering pipelines, and low-latency model inference services using FastAPI.
- ▶ Implemented MLflow-based model tracking, performance monitoring, and drift detection to support continuous model improvement.
- ▶ Containerized services using Docker and developed scalable deployment architecture for production ML workflows.

Multimodal Document Intelligence System | PyTorch, Transformers, OCR, RAG, Vector Database

- ▶ Developed an AI-powered document understanding platform combining computer vision, OCR, NLP, and large language models for intelligent document analysis.
- ▶ Built a Retrieval-Augmented Generation (RAG) pipeline enabling semantic search and question answering over unstructured documents.

- ▶ Integrated transformer-based models for document classification, information extraction, and contextual summarization.
- ▶ Optimized inference workflows using vector embeddings and similarity search for efficient knowledge retrieval.

PROFESSIONAL EXPERIENCE

Software Developer | *HCL Technologies* | India

May 2023 – May 2024

- ▶ Designed and maintained 6-table normalized PostgreSQL database for structured operational data; wrote complex multi-table JOIN queries, subqueries, and aggregations across 1M+ record datasets.
- ▶ Engineered automated Python (Pandas, NumPy) ETL pipelines reducing manual preprocessing effort by 40%; enforced full analytical reproducibility across varied dataset schemas.
- ▶ Developed ML classification and anomaly detection models using scikit-learn; improved F1-score by 18% through structured cross-validation and hyperparameter tuning across 5+ model variants.
- ▶ Applied NLP (TF-IDF, NER, tokenization) to extract structured insights from unstructured text corpora, reducing analyst review time by 30%.
- ▶ Deployed scalable microservices on AWS (EC2, S3) with CI/CD pipelines, achieving 99.9% production uptime across all releases.

Encrypted File Chunking System | *Python, AES-256, Cloud Infrastructure, Metadata Management*

- ▶ Engineered secure file encryption and chunking architecture for cloud storage, published at ICET 2024 (12th International Conference on Contemporary Engineering and Technology).
- ▶ Implemented AES-256 encryption combined with adaptive chunking algorithm on 5,000+ test files ranging from 100 MB to 10 GB; achieved 256-bit cipher strength with zero data loss across distributed cloud nodes.
- ▶ Designed metadata management module tracking 8 encryption parameters per file (algorithm, key derivation, chunk mapping, access logs) with 99.97% retrieval accuracy.
- ▶ Demonstrated governance compliance with GDPR/HIPAA metadata requirements; documented findings in peer-reviewed publication highlighting 15% faster encryption throughput vs. baseline and 12% storage efficiency gain through deduplication.

EDUCATION

M.S. Computer Science | *University of Florida* | Gainesville, FL

Aug 2024 – May 2026

GPA: 3.79 / 4.0

B.E. Computer Science | *Sathyabama University* | Chennai, India

Aug 2020 – May 2024

GPA: 8.78 / 10.0

CERTIFICATIONS & RECOGNITION

- ▶ Survival Analysis in R for Public Health — Imperial College London, - Kaplan-Meier curves, Cox Proportional Hazards models, censoring, hazard functions applied to public health data
- ▶ A Crash Course in Causality: Inferring Causal Effects from Observational Data - University of Pennsylvania, Coursera — DAGs, propensity score matching, instrumental variables, inverse probability of treatment weighting
- ▶ Machine Learning with Scikit-learn, PyTorch & Hugging Face Professional Certificate - Coursera (- supervised/unsupervised learning, time series, NLP, generative models
- ▶ Exam Prep (NCA-GENL): NVIDIA-Certified Generative AI LLMs Specialization — Whizlabs — LLM architecture, NLP, AI deployment, prompt engineering
- ▶ NVIDIA: Prompt Engineering and Data Analysis — Whizlabs NVIDIA: LLM Experimentation, Deployment, and Ethical AI — Whizlabs
- ▶ Advanced Machine Learning Techniques